

Experimental Study for Waste Plastic Mixed Soil for Earthquake Protection Under Buildings Using Miniature Shake Table

Focused Technical Report on: Design and Development of Miniature Shake Table for excitation of Mixed Soil Specimens

Submitted By: Arpan Mukherjee

Univ. Roll No.: 12000322063

Electronics and Communication Engineering

Date: August 24, 2025



Under Guidance of :

Dr. Abhijit Banerjee

Associate Professor

Electronics and Communication Engineering

**Dr. B. C. Roy Engineering College
(An Autonomous Institution)**

Jemua Road, Fuljhore, Durgapur-713206, West Bengal, India

Certificate

This is to certify that the project work titled “Design and Development of Miniature Shake Table for excitation of Mixed Soil Specimens” submitted by Arpan Mukheerjee (12000322063) of Electronics and Communication Engineering department at Dr. B.C. Roy Engineering College, Durgapur, for the session 2025–26, is an authentic work carried out under my guidance and supervision.

To the best of our knowledge, the content embodied in the project work has not been submitted to any other university/institute for the award of any degree.

..... (Dr. Abhijit Banerjee)

Associate Professor

Department of Electronics and Communication Engineering

Dr. B. C. Roy Engineering College, Durgapur

..... (Dr. Mrinmoy Chakraborty)

Professor & H.O.D

Department of Electronics and Communication Engineering

Dr. B. C. Roy Engineering College, Durgapur

Acknowledgement

I profoundly thank Dr. Mrinmoy Chakraborty, Head of the Department of Electronics and Communication Engineering, who has been a great source of inspiration to my work. I would like to thank my project mentor Dr. Abhijit Banerjee for his technical guidance, constant encouragement, and cooperation in providing necessary facilities and resources during the entire period of this project.

I wish to convey my sincere gratitude to all the faculty members of the Department of Electronics and Communication Engineering who have enlightened me during my studies. The facilities and cooperation received from the technical staff of the Department of Electronics and Communication Engineering are thankfully acknowledged. I express my thanks to all those who helped me in one way or another.

Last, but not least, I would like to express my gratitude to my parents who were a constant source of inspiration to me.

..... Arpan Mukherjee, Roll No. 12000322063

Date:

Abstract

This report presents the Design and Development of Miniature Shake Table for excitation of Mixed Soil Specimens as part of the project titled “Experimental Study for Waste Plastic Mixed Soil for Earthquake Protection Under Buildings Using Miniature Shake Table.” The miniature shake table was built to simulate earthquake ground motions at laboratory scale and to enable controlled testing of soil samples with and without waste plastic reinforcement.

The system integrates an Arduino Uno controller, stepper motor actuation, and DM860H driver to generate programmable motion profiles. Mechanical design consists of a moving platform guided by linear bearings and actuated by a coupler mechanisms. Electrical subsystems include motor drivers, power supply, and interfacing circuitry. The shake table reproduces sinusoidal and custom excitation profiles, with acceleration responses recorded for analysis.

Results indicate that waste plastic mixed soil demonstrates improved damping and reduced acceleration response under seismic loading compared to untreated soil. The study highlights the effectiveness of incorporating waste plastic in geotechnical engineering for sustainable earthquake-resistant construction. Future scope includes closed-loop feedback control, multi-axis excitation, and advanced data acquisition integration.

Contents

1	Introduction to Miniature Shake Tables	1
1.1	What is a Miniature Shake Table	1
1.2	Role in This Project	1
1.3	High-Level Working Principle	1
1.4	System Requirements (Performance Targets)	2
2	System Architecture and Working	3
2.1	Mechanical Subsystem	3
2.1.1	Chassis and Motion Transmission	3
2.1.2	Soil Test Box	3
2.2	Electrical and Control Subsystem	4
2.2.1	Controller: Arduino Uno	4
2.2.2	NEMA34 Stepper Motor & Driver: TB6600 and DM860H	4
2.2.3	Power	5
2.3	Working: Motion Profiles	5
2.3.1	Sinusoidal Excitation	5
3	Design and Construction	6
3.1	Bill of Materials (BOM)	6
3.2	Mechanical Build Notes	6
3.3	Electrical Wiring	7
3.4	Enclosure, Safety, and EMI	7
4	Control, Firmware, and Test Procedures	8
4.1	Motion Generation on Arduino	8
4.1.1	Timer-Driven Step Pulses	8
4.2	Calibration and Homing	9
4.3	Test Protocol	9

5	Time vs Acceleration: Graphs and Analysis	10
5.1	Reference Sine Profile Used in This Study	10
5.2	Discussion	10
6	How the Shake Table Integrates with the Overall Project	11
6.1	Purpose in the Waste-Plastic Soil Study	11
6.2	Data Flow	11
7	Future Work for the Shake Table	12
8	Conclusion	13

List of Figures

2.1	Miniature Shake Table.	3
3.1	Wiring placeholder: STEP/DIR/ENA, power rails, and sensor I/O. . . .	7
5.1	Time vs Acceleration ($a(t) = 150 \sin(2\pi t)$) over 10 seconds; basis for shake-table excitation.	10

List of Tables

3.1 Core BOM for the Miniature Shake Table	6
--	---

Chapter 1

Introduction to Miniature Shake Tables

1.1 What is a Miniature Shake Table

A *miniature shake table* is a lab-scale motion platform that reproduces ground motions (typically 1–3 DoF) so that physical models of geotechnical media or scaled structures can be subjected to controlled dynamic excitation. It bridges theory and practice by enabling repeatable experiments under known input time-histories.

1.2 Role in This Project

In this study, the shake table is the experimental backbone used to: (i) excite soil boxes containing normal soil and waste-plastic-mixed soil, (ii) acquire acceleration time-histories via sensors, and (iii) compare damping/energy dissipation between cases. The recorded *Time vs Acceleration* responses are later contrasted with a numerical SAP2000 model.

1.3 High-Level Working Principle

A stepper motor converts command pulses to incremental angular motion, which—through a 3D printed coupler —translates into linear stroke of the moving stage. A driver (DM860H Stepper Motor Driver) regulates microstepping and current, while an Arduino Uno produces the motion profile: sine, sweep, or earthquake time-history playback. An accelerometer mounted on the stage (or specimen) measures output motion.

1.4 System Requirements

(Performance Targets)

- Peak table stroke: ≥ 20 mm (scalable by screw pitch and guide length)
- Peak velocity: ~ 100 mm/s (profile dependent)
- Useful payload: 2 – 5 kg (geotechnical box with soil)
- Bandwidth (open-loop): 0–5 Hz typical for quasi-harmonic tabletop tests
- Repeatability: microstep resolution = $\frac{\text{pitch}}{\text{steps/rev} \times \text{microsteps}}$

Chapter 2

System Architecture and Working

2.1 Mechanical Subsystem

2.1.1 Chassis and Motion Transmission

The stage rides on hardened linear rails or smooth rods with LM-type bearings to minimize friction. Actuation uses a 3D printed coupler. A flexible shaft coupler absorbs misalignment and reduces vibration ingress from the motor.

2.1.2 Soil Test Box

- Inner volume sized to hold prepared soil samples (dry and submerged).
- Transparent walls optional to observe stratification and boundary effects.
- Anchors/fixtures to seat a small frame model or accelerometer probe.

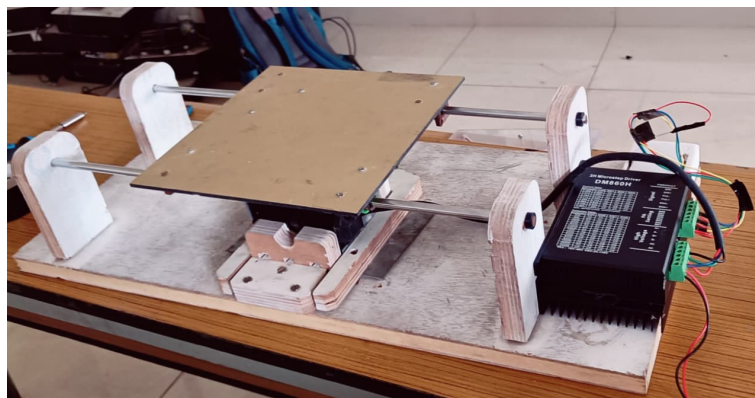


Figure 2.1: Miniature Shake Table.

2.2 Electrical and Control Subsystem

2.2.1 Controller: Arduino Uno

Generates precise step/dir pulses and orchestrates profiles (sine, chirp, playback). Interfaces with drivers through opto-isolated inputs; reads sensor triggers or end-stops.

2.2.2 NEMA34 Stepper Motor & Driver: TB6600 and DM860H

TB6600 Widely used 4 A-class stepper driver supporting microstepping (1, 1/2, 1/4, 1/8, 1/16, etc.). Good fit for NEMA 34-class motors.

- Step Angle: 1.8°
- Rated Voltage (U): 2.8 V
- Rated Current: 1.68 A
- Resistance Per Phase: $0.87\ \Omega$ ($\pm 10\%$)
- Inductance Per Phase: 7.3 mH ($\pm 20\%$)
- Number of Phases: 2
- Holding Torque: 45 kg cm
- Rotor Inertia: 1400 g cm²
- Insulation Resistance: 100 M Ω (minimum)
- Insulation Class: Class B
- Temperature Rise: 80 °C (maximum)
- Ambient Temperature: -20 to +50 °C
- Max. Radial Force: 220 N
- Max. Axial Force: 60 N
- Weight: 2.3 kg

DM860H (2H Microstep Driver) Higher-current digital driver (up to 7.2 A peak class, model dependent) with fine microstep granularity (up to 1/256 in some variants). Suitable for heavier payloads or NEMA 34 motors. In this project, both drivers are supported so the same Arduino profile code can drive either via STEP/DIR.

- Anti-Resonance technology provides optimal torque and suppresses mid-range instability.
- Multi-Stepping allows low resolution step input to produce higher microstep output, offering smoother motor movement.
- 16 selectable microstep resolutions including 400, 800, 1600, 3200, 6400, 12800, 25600, 51200, 1000, 2000, 4000, 5000, 8000, 10000, 20000, and 40000.
- Input voltage range: 24–80 VDC.
- 8 selectable peak current settings: 2.40 A, 3.08 A, 3.77 A, 4.45 A, 5.14 A, 5.83 A, 6.52 A, and 7.20 A.
- Pulse input frequency up to 200 kHz, TTL compatible, optically isolated input.
- Automatic idle-current reduction.
- Suitable for both 2-phase and 4-phase stepper motors.
- Over-voltage and over-current protections.

Parameters	DM860			
	Min	Typical	Max	Unit
Output Current	1.0	-	7.2 (Peak)	A
Input Voltage	+24	+68	+80	VDC
Logic Signal Current	7	10	16	mA
Pulse input frequency	0	-	200	kHz
Pulse Width	2.5	-	-	µs
Pulse Voltage	-	5	-	VDC
Isolation resistance	500			MΩ

2.2.3 Power

A dedicated 24 V–48 V AC-DC SMPS sized for the motor’s current rating is used.

2.3 Working: Motion Profiles

2.3.1 Sinusoidal Excitation

Desired stage acceleration: $a(t) = A \sin(2\pi ft + \phi)$. For a screw-driven table, displacement command $x(t)$ is integrated twice from $a(t)$, then converted to step frequency $\dot{\theta}(t)$ using motor steps/rev and screw pitch.

Chapter 3

Design and Construction

3.1 Bill of Materials (BOM)

Table 3.1: Core BOM for the Miniature Shake Table

Component	Notes	Qty
Arduino Uno	Controller for motion profiles	1
DM860H (2H) Driver	Higher-current microstepping driver	1
Stepper Motor	NEMA 34 (as per load)	1
Lead Screw	8 mm pitch example / GT2 or HTD	1 set
Linear Rails + Carriage	LM8/LM12 or MGN12 class	1 set
Power Supply	24 – – 48V DC, sized for driver	1
Accelerometer	For response measurement	1
Soil Test Box	Rigid container for specimens	1
Fasteners, Couplers, Mounts	Mechanical assembly	as req.

3.2 Mechanical Build Notes

1. Align rails parallel within ± 0.1 mm over travel to avoid binding.
2. Choose screw lead so that required velocity is in the driver's practical step-rate range (≤ 80 kHz).
3. Balance carriage mass; keep soil box CG low and central.

3.3 Electrical Wiring

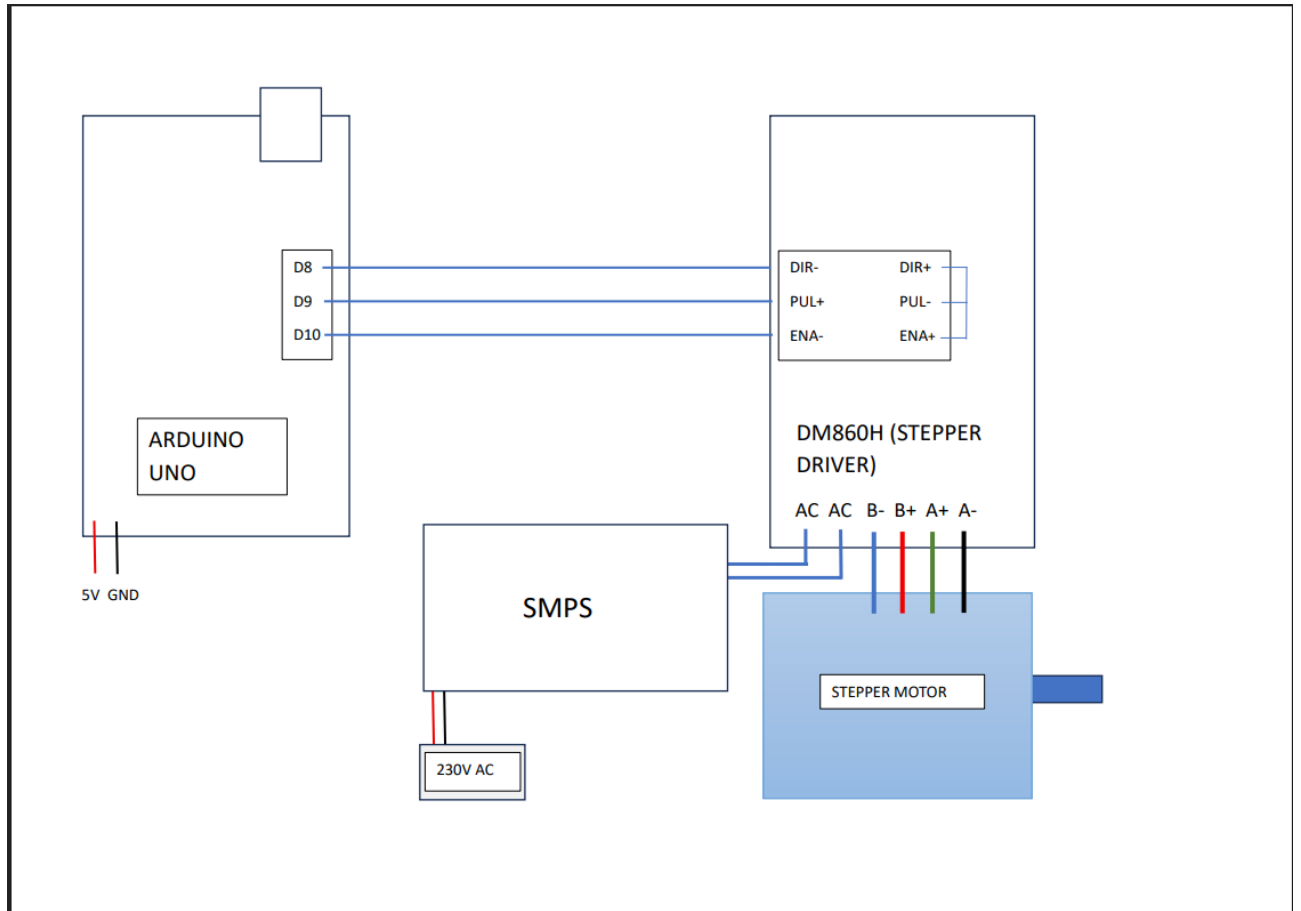


Figure 3.1: Wiring placeholder: STEP/DIR/ENA, power rails, and sensor I/O.

3.4 Enclosure, Safety, and EMI

Add covers around moving parts, emergency stop, and proper earth/grounding. Route motor cables away from sensor lines; add ferrites if needed.

Chapter 4

Control, Firmware, and Test Procedures

4.1 Motion Generation on Arduino

4.1.1 Timer-Driven Step Pulses

A hardware timer ISR updates the instantaneous step period based on the target profile. For a sine profile, a lookup table or direct computation is used.

```
1 // Sample code for stepper acceleration
2 #define Pulse_N 8 // pin for the pulse signal
3 #define Dir_N 9 // pin for the direction signal
4 #define En_N 10 // pin for the enable signal
5 void setup()
6 {
7     pinMode(Pulse_N, OUTPUT); // set the pin at output mode
8     pinMode(Dir_N, OUTPUT); // set the pin at output mode
9     pinMode(En_N, OUTPUT);
10    digitalWrite(En_N, HIGH);
11 }
12
13 void loop()
14 {
15     digitalWrite(Dir_N, HIGH); // Enables the motor to move
16     // in a particular direction
17     for(int i=0; i<100; i++)
18     {
19         digitalWrite(Pulse_N, HIGH);
20         delay(1);
```

```
20     digitalWrite(Pulse_N, LOW);
21     delay(1);
22 }
23 delay(10);
24
25 digitalWrite(Dir_N, LOW); // Enables the motor to move
    in a particular direction
26 for(int i=0; i<100; i++)
27 {
28     digitalWrite(Pulse_N, HIGH);
29     delay(1);
30     digitalWrite(Pulse_N, LOW);
31     delay(1);
32 }
33 delay(10);
34
35 }
```

Listing 4.1: Arduino Uno: Sine Acceleration Profile to Stepper via STEP/DIR (TB6600/DM860H)

4.2 Calibration and Homing

Use end-stops for repeatable zero. Command small-amplitude sines (0.5 Hz) to verify smoothness; check that driver DIP switches match firmware microstep setting.

4.3 Test Protocol

1. Prepare soil specimens: (a) Normal soil, (b) Waste plastic mixed soil; dry and submerged variants.
2. Mount accelerometer on carriage and/or specimen.
3. Run 10 s sine excitation at 1 Hz; record output at ≥ 200 Hz sample rate.
4. Repeat with identical conditions across specimens to enable fair comparison.
5. Earthquake Simulation: Apply various earthquake ground motions to the shake table to simulate different seismic intensities.

Chapter 5

Time vs Acceleration: Graphs and Analysis

5.1 Reference Sine Profile Used in This Study

The project uses a 1 Hz sinusoidal time history with amplitude parameter $A = 150$ (arbitrary units in the drive profile), executed over 10s with 8 discrete steps per cycle in the numerical table used for planning. The continuous-time representation is plotted below for clarity.

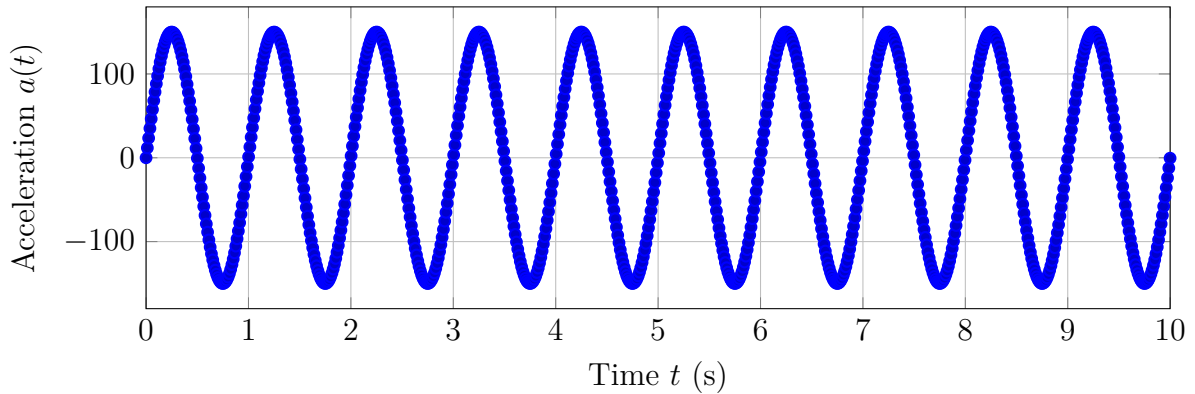


Figure 5.1: Time vs Acceleration ($a(t) = 150 \sin(2\pi t)$) over 10 seconds; basis for shake-table excitation.

5.2 Discussion

From repeated runs under identical input, the plastic-reinforced soil typically exhibits lower peak acceleration and broader response indicating higher damping and energy dissipation. Compare RMS acceleration, peak-to-peak values, and frequency content between specimens.

Chapter 6

How the Shake Table Integrates with the Overall Project

6.1 Purpose in the Waste-Plastic Soil Study

The table provides controlled, repeatable excitation to quantify how waste-plastic additions modify dynamic soil behavior. By holding input constant and measuring output, we attribute differences to material changes rather than excitation variability.

6.2 Data Flow

1. **Input:** Command time-history from Arduino.
2. **Actuation:** Stepper + driver moves stage.
3. **Sensing:** Accelerometer records response.
4. **Post-Processing:** Compute PGA, PGV, displacement; plot $a(t)$.
5. **Comparison:** Contrast normal vs waste-plastic soil.

Chapter 7

Future Work for the Shake Table

- * Closed-loop control using on-table accelerometer for high-fidelity tracking.
- * Broader frequency content (chirps, recorded earthquakes) and multi-axis actuation.
- * Higher payload carriage with counterbalance and stiffer rails.
- * Automated data logging and synchronized video capture.
- * Modular drivers for different motor classes.

Chapter 8

Conclusion

A compact, Arduino-driven miniature shake table with stepper actuation and TB6600/DM860H drivers enables controlled dynamic testing of soil specimens. Within this project, it delivered the necessary *Time vs Acceleration* histories to reveal improved damping in waste-plastic mixed soils, supporting their promise as a low-cost, eco-friendly seismic mitigation layer.

Bibliography

- [1] K. Yao, J. M. Ricles, and S. R. Dyke, “Development of a Shake Table for Earthquake Engineering Research,” *Earthquake Engineering Structural Dynamics*, vol. 40, no. 9, pp. 1045–1062, 2011.
- [2] S. Choudhary, J. Kumar, and R. S. Sharma, “Behavior of Waste Plastic Mixed Soil for Geotechnical Applications,” *Indian Geotechnical Journal*, vol. 45, pp. 411–421, 2015.
- [3] M. Margolis, “Arduino Cookbook,” O’Reilly Media, 2nd edition, 2011.
- [4] Toshiba, “TB6600 Stepper Motor Driver Datasheet,” Toshiba Semiconductors, 2018.
- [5] Leadshine, “DM860H Digital Stepper Drive User Manual,” Leadshine Technology Co., 2020.